

Extender Algebra

Obrad Kasum

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1 Infinitary Logic

- I. **Definition.** Let δ be an inaccessible cardinal and let $\mathbf{P} \subseteq H_\delta$ be an infinite set of propositional letters. The formulas of the propositional calculus $\mathcal{L}_\delta(\mathbf{P})$ are determined as follows.
 - a. $(p, ())$ is a formula for every $p \in \mathbf{P}$.
 - b. $(\perp, ())$ and $(\top, ())$ are formulas.
 - c. $(\neg, (\phi))$, $(\Leftarrow, (\phi, \psi))$, $(\Leftrightarrow, (\phi, \psi))$ are formulas, for all formulas ϕ, ψ .
 - d. $(\wedge, (\phi_\xi : \xi < \gamma))$ and $(\vee, (\phi_\xi : \xi < \gamma))$ are formulas, for all $\gamma < \delta$ and all formulas ϕ_ξ ($\xi < \gamma$).
 - e. The set of formulas is minimal with respect to the previous rules.
- II. **Remark.** All the formulas are contained in H_δ . The formulas, viewed as trees, are well-founded.
- III. **Definition.** The rank of a formula is the rank of the tree that corresponds to it.
- IV. **Remark.** It is possible to do induction on the rank of a formula.
- V. **Definition.** A theory of $\mathcal{L}_\delta(\mathbf{P})$ is a set of formulas of that language.
- VI. **Definition.** The set of proofs in $\mathcal{L}_\delta(\mathbf{P})$ is determined as follows.
 - a. The triple $(\text{hypo}, (), \emptyset, \phi)$ is a proof, where ϕ is a formula.
 - b. If π, ρ, σ are proofs, then so is any triple of the form:
 - i. the first coordinate is a rule of natural deduction which is not $\wedge_i$, $\wedge_e$, $\vee_i$ or $\vee_e$;
 - ii. the second coordinate is a 1-tuple, pair or triple of proofs π, ρ, σ , depending on the arity of the rule from (a);
 - iii. the third coordinate is the set of paths through the tree towards the discharged hypotheses;
 - iv. the fourth coordinate is the conclusion of the rule.

- c. If π_ξ are proofs for $\xi < \gamma < \delta$, then so is any triple of the form¹:
 - i. the first coordinate is one of the rules $\wedge_i, \wedge_e, \vee_i, \vee_e$;
 - ii. the second coordinate is $(\pi_\xi : \xi < \gamma)$;
 - iii. the third coordinate is the set of paths through the tree towards the discharged hypotheses;
 - iv. the fourth coordinate is the conclusion of the rule.
 - d. The set of proofs is minimal with respect to a-c.
- VII. **Remark.** The set of all proofs is contained in H_δ . The proofs, viewed as trees, are well-founded.
- VIII. **Definition.** The rank of a proof is the rank of the tree that corresponds to it.
- IX. **Remark.** As in the case of formula, one can to proofs by induction on the rank of a proof.
- X. **Definition.** A proof in a theory T of $\mathcal{L}_\delta(\mathbf{P})$ is a proof in $\mathcal{L}_\delta(\mathbf{P})$ all of whose non-discharged hypotheses are in T . $T \vdash \phi$ if there is a proof in T whose root has ϕ for its fourth coordinate.
- XI. If $T \vdash \phi$, then there is $T' \in [T]^{<\delta}$ such that $T' \vdash \phi$.

Proof.

- 1° Let π witness $T \vdash \phi$.
- 2° $\pi \in H_\delta$
- 3° Thus, the $T' \subseteq T$ of all the non-discharged hypotheses of π belongs to H_δ .
- 4° Thus, $T' \in [T]^{<\delta}$.
- 5° π witnesses $T' \vdash \phi$.

□

- XII. **Definition.** An evaluation is a mapping $\mu : \mathbf{P} \rightarrow 2$. For a formula ϕ of $\mathcal{L}_\delta(\mathbf{P})$, the evaluation $\mu(\phi)$ is defined in the obvious manner by induction on the rank of ϕ .
- XIII. **Remark.** To compute $\mu(\phi)$, only the propositional letters appearing in ϕ are relevant.
- XIV. **Definition.** Let ϕ be a formula and T be a theory of $\mathcal{L}_\delta(\mathbf{P})$; let μ be an evaluation.

$$\begin{aligned}
 \mu \models \phi &: \iff \mu(\phi) = 1, \\
 \mu \models T &: \iff (\forall \psi \in T)(\mu \models \psi), \\
 T \models \phi &: \iff (\forall \nu \in 2^{\mathbf{P}})(\nu \models T \implies \nu \models \phi).
 \end{aligned}$$

¹here we describe the infinite versions of the usual rules for $\wedge$ and $\vee$

XV. Proposition.

- a. If $T \vdash \phi$, then $T \models \phi$.
- b. If $\mathbf{P} \in H_\delta$, then there is ϕ such that $\models \phi$, but $\nVdash \phi$.

Proof.

- 1° Point a. follows by induction on the rank of ϕ .
- 2° We work now on the point b.
- 3° Let us enumerate $\mathbf{P} = \{p_\xi : \xi < \kappa\}$, where $\kappa < \delta$.
- 4° Let us enumerate $2^\mathbf{P} = \{\mu_\eta : \eta < \lambda\}$, where $\lambda := 2^\kappa < \delta$.
- 5° We use notation $p_\xi^1 := p_\xi$ and $p_\xi^0 := \neg p_\xi$.
- 6° Set

$$\phi := \bigvee_{\eta < \lambda} \bigwedge_{\xi < \kappa} p_\xi^{\mu_\eta(\xi)}.$$

- 7° Since every evaluation is enumerated, we have $\models \phi$.
- 8° Let $g : \mathbf{P} \rightarrow 2$ be V -generic for $\text{Add}(1, \kappa)$.
- 9° We note that this poset preserves the inaccessibility of δ .
- 10° Since $g \neq \mu_\eta$ for every $\eta < \lambda$, we have

$$V[g] \models (\neq \phi).$$

- 11° The point 1 yields

$$V[g] \models (\nVdash \phi).$$

- 12° Consequently,

$$V \models (\nVdash \phi).$$

□

XVI. Proposition.

- a. If $T \vdash \phi$, then $(\forall \mathbb{P} \in H_\delta)(V^\mathbb{P} \models (T \models \phi))$.
- b. $(\forall \mathbb{P} \in H_\delta)(V^\mathbb{P} \models (\models \phi)) \implies \vdash \phi$.
- c. $(\forall \kappa < \delta)(V^{\text{Col}(\omega, \kappa)} \models (\models \phi)) \implies \vdash \phi$.

Proof.

- 1° For a, we just apply the previous proposition in $V^\mathbb{P}$.
- 2° Point b. follows from point c.
- 3° We now work on c.
- 4° Assume that $\nVdash \phi$. We want to find $\kappa < \delta$ and $[\text{in } V^{\text{Col}(\omega, \kappa)}]$ an evaluation $\mu \models \neg \phi$.

- 5° We may assume w.l.o.g. that ϕ only uses $\neg$ and $\wedge$.
- 6° Let $\{\psi_\xi : \xi < \kappa\}$ be the set of all subformulas of $\neg\phi$, with $\psi_0 = \neg\psi$, where $\kappa < \delta$.
- 7° Let $\mathbf{Q}$ be the set of all propositional letters of ϕ .
- 8° We work in $V^{\text{Col}(\omega, \kappa)}$. Let $g : \omega \rightarrow \kappa$ be the generic.
- 9° $(T_n : 0 < n < \omega)$ is defined by recursion.
 - 1' $T_1 := \{\neg\phi\}$.
 - 2' Suppose that T_n is given for some $n > 0$.
 - 1'' In the case $T_n \vdash \psi_{g(n)}$, we set $T_{n+1} := T_n \cup \{\psi_{g(n)}\}$.
 - 2'' In the case $T_n \not\vdash \psi_{g(n)}$, we set $T'_n := T_n \cup \{\neg\psi_{g(n)}\}$.
 - 1''' In the subcase $\psi_{g(n)} \in \mathbb{P}$, we set $T_{n+1} := T'_n$.
 - 2''' In the subcase $\psi_{g(n)} = \neg\theta$ for some θ , we set $T_{n+1} := T'_n \cup \{\theta\}$.
 - 3''' In the subcase $\psi_{g(n)} = \bigwedge_{\xi < \gamma} \theta_\xi$ for some $(\theta_\xi : \xi < \gamma)$, there is $\xi < \gamma$ such that $T_n \not\vdash \theta_\xi$. We set $T_{n+1} := T_n \cup \{\neg\theta_\xi\}$.
- 10° It is shown by induction that every T_n is a finite consistent set of formulas in V .
- 11° Also, $(T_n : 0 < n < \omega)$ is increasing.
- 12° $T := \bigcup_{0 < n < \omega} T_n$
- 13° For $p \in \mathbf{Q}$, we define

$$\mu(p) := \begin{cases} 1, & p \in T \\ 0, & \neg p \in T. \end{cases}$$

It is clear that the definition is correct.

- 14° For every $\xi < \kappa$, exactly one holds: $\psi_\xi \in T$ or $\neg\psi_\xi \in T$.
- 15° For every $\xi < \kappa$, we have $\mu(\psi_\xi) = 1$ if and only if $\psi_\xi \in T$.

Proof.

- 1' This is shown by induction on the rank of ψ_ξ .
- 2' The case of propositional letters holds by the definition.
- 3' Assume that $\psi_\xi = \neg\psi_\eta$ and $\mu(\psi_\xi) = 1$.
- 4' This means $\mu(\psi_\eta) = 0$.
- 5' By IH, $\psi_\eta \notin T$.
- 6' Let $i : g(i) = \eta$.
- 7' $T_i \not\vdash \psi_\eta$
- 8' $\neg\psi_\eta \in T_{i+1}$
- 9' Thus, $\psi_\xi \in T$.
- 10' Assume now that $\psi_\xi = \neg\psi_\eta$ and $\mu(\psi_\xi) = 0$.
- 11' This means $\mu(\psi_\eta) = 1$.
- 12' Thus, $\psi_\eta \in T$.
- 13' Thus, $\neg\psi_\eta \notin T$.

- 14' Thus, $\psi_\xi \notin T$.
 15' Assume now that $\psi_\xi = \bigwedge_{\alpha < \gamma} \psi_{\eta_\alpha}$ and $\mu(\psi_\xi) = 1$.
 16' $(\forall \alpha < \gamma)(\mu(\psi_{\eta_\alpha}) = 1)$
 17' $(\forall \alpha < \gamma)(\psi_{\eta_\alpha} \in T)$
 18' $(\forall \alpha < \gamma)(\neg \psi_{\eta_\alpha} \notin T)$
 19' Thus, the case $9^\circ 2' 2'' 3'''$ cannot occur for $n : g(n) = \xi$.
 20' Thus, $T_n \vdash \psi_\xi$.
 21' Thus, $\psi_\xi \in T_{n+1} \subseteq T$.
 22' Assume now that $\psi_\xi = \bigwedge_{\alpha < \gamma} \psi_{\eta_\alpha}$ and $\mu(\psi_\xi) = 0$.
 23' This means $\mu(\psi_{\eta_\alpha}) = 0$ for some $\alpha < \gamma$.
 24' $\psi_{\eta_\alpha} \notin T$
 25' $\neg \psi_{\eta_\alpha} \in T$
 26' If it were the case that $\psi_\xi \in T$, we would have $\psi_\xi, \neg \psi_{\eta_\alpha} \in T_m$ for some m , yielding the inconsistency of T_m .
 27' Thus, $\psi_\xi \notin T$.

□

16° Since $\neg \phi \in T$, we conclude $\mu(\phi) = 0$.

□

XVII. We denote by $\mathbb{B}_{\delta, \mathbf{P}}$ the Lindenbaum algebra of $\mathcal{L}_\delta(\mathbf{P})$. Recall that its elements are the equivalence classes $[\phi]$ of the equivalence relation $\sim$ defined by:

$$\phi \sim \psi : \Longleftrightarrow \vdash_{\mathcal{L}_\delta(\mathbf{P})} \phi \leftrightarrow \psi.$$

This algebra is δ -complete.

XVIII. For a theory T of $\mathcal{L}_\delta(\mathbf{P})$, we define

$$F_T := \{[\phi] : T \vdash \phi\} \subseteq \mathbb{B}_{\delta, \mathbf{P}}.$$

It is easily seen that F_T is a δ -complete filter. We define $\mathbb{B}_{\delta, \mathbf{P}}(T) := \mathbb{B}_{\delta, \mathbf{P}}/F_T$, the Lindenbaum algebra of T . This is a δ -complete boolean algebra. One can also understand it as the set of equivalence classes $[\phi]_T$ of formulas modulo $\sim_T$, where

$$\phi \sim_T \psi : \Longleftrightarrow T \vdash \phi \leftrightarrow \psi.$$

XIX. A theory T is said to be κ -saturated if F_T is κ -saturated. In this case, $\mathbb{B}_{\delta, \mathbf{P}}(T)$ is κ -c.c. In particular, if T is δ -saturated, then $\mathbb{B}_{\delta, \mathbf{P}}(T)$ is complete.

XX. Suppose that:

- M is a transitive model of set theory;
- In M :

- δ is inaccessible;
- $\mathbf{P} \subseteq H_\delta$
- T is a theory of $\mathcal{L}_\delta(\mathbf{P})$;
- $\mu : \mathbf{P} \rightarrow 2$.

We define $F_\mu := \{[\phi]^M : \mu \models \phi\}$. This is an ultrafilter on $\mathbb{B}_{\delta, \mathbf{P}}$ and it is M - δ -complete. Notice that, given a theory T , we have $F_\mu \supseteq F_T$ if and only if $\mu \models T$. In this case, we also have an ultrafilter $F_\mu^T := \{[\phi]_T^M : \mu \models \phi\}$ on $\mathbb{B}_{\delta, \mathbf{P}}(T)$.

XXI. Suppose that:

- M is a transitive model of set theory;
- In M :
 - δ is inaccessible;
 - $\mathbf{P} \subseteq H_\delta$
 - T is a theory of $\mathcal{L}_\delta(\mathbf{P})$;
 - T is δ -saturated;
- $\mu : \mathbf{P} \rightarrow 2$;
- $\mu \models T$.

Then $F_\mu^T \rightsquigarrow M^{\mathbb{B}_{\delta, \mathbf{P}}(T)}$.

Proof.

1° In M , let A be a maximal anti-chain of $\mathbb{B}_{\delta, \mathbf{P}}(T)$. We want to show $A \cap F_\mu^T \neq \emptyset$.

2° We have

$$\left[\bigvee_{[\phi]_T \in A} \phi \right] = \sum A = 1 \in F_\mu^T.$$

3° This implies

$$\mu \models \bigvee_{[\phi]_T \in A} \phi.$$

4° Thus, $\mu \models \phi$ for some $\phi : [\phi]_T \in A$.

5° Consequently, $[\phi]_T \in A \cap F_\mu^T$.

□

2 Iteration trees

2.1 An Important Lemma

I. Suppose that:

- a. M is a countable transitive model of set theory;
- b. $\mathcal{T}$ is an iteration on M of length $\omega_1 + 1$;
- c. $b := \{\xi : \xi \triangleleft^{\mathcal{T}} \omega_1\}$;
- d. $k : (H, \in, \mathcal{U}, \alpha, c) \prec (H_{\omega_2}, \in, \mathcal{T}, \omega_1, b)$ is a countable hull.

Then:

- i. $\text{crit}(k) = \alpha$
- ii. $\mathcal{U} \restriction \alpha = \mathcal{T} \restriction \alpha$
- iii. $c = b \cap \alpha$
- iv. $\alpha \in b$
- v. $c = \{\xi : \xi \triangleleft^{\mathcal{U}} \alpha\} = \{\xi : \xi \triangleleft^{\mathcal{T}} \alpha\}$
- vi. $\mathcal{U} \triangleleft \mathcal{T}$
- vii. $k \restriction M_\alpha = j_{\alpha, \omega_1}$

Proof.

1° i. is obvious.

2° In particular, $\mathcal{U} \restriction \alpha = k[\mathcal{U} \restriction \alpha] = \mathcal{T} \restriction \alpha$.

3° $b \cap \alpha = j(c) \cap j[\alpha] = j[c \cap \alpha] = c$

4° $\omega_1 = \sup b$

5° Thus, $\alpha = \sup c$.

6° Consequently, $\alpha = \sup(b \cap \alpha)$.

7° Since b is club, we have $\alpha \in b$.

8° v. follows from c. by the elementarity.

9° vi. follows from ii. and v.

10° Let $x \in M_\alpha$ be arbitrary. We want to show that $k(x) = j_{\alpha, \omega_1}(x)$.

11° We have $x = j_{\beta, \alpha}(y)$ for some $\beta < \alpha$ and $y \in M_\beta$.

12° This means $k(x) = k(j_{\beta, \alpha}(y)) = j_{k(\beta), k(\alpha)}(k(y)) = j_{\beta, \omega_1}(y) = j_{\alpha, \omega_1}(j_{\beta, \alpha}(y)) = j_{\alpha, \omega_1}(x)$.

□

2.2 $\mathcal{E}$ -iterations

II. Let M be a transitive model of set theory and let $\mathcal{E}$ be such that in M :

- $\mathcal{E}$ is a set of extenders;
- $\text{strength}(E) = \text{length}(E)$ is inaccessible.

An $\mathcal{E}$ -iteration (tree) on M of length α is an iteration (tree) (in the sense of the mémoire) on M satisfying

$$(\forall \xi < \alpha)(E_\xi \in j_{0,\xi}(\mathcal{E})).$$

III. An $\mathcal{E}$ -iteration game is the same as the iteration game, with additional requirement that an extender E_ξ that I picks at some stage $\xi + 1$ must belong to $j_{0,\xi}(\mathcal{E})$. The notions of iteration strategy and iterability are adjusted accordingly.

3 Extender Algebra

I. **Definition.** Let δ be inaccessible and $f : \delta \rightarrow H_\delta$. Then $\text{strong}(E, \kappa, \lambda, f)$ holds if:

- a. $\lambda < \delta$ is limit;
- b. E is a strong (κ, λ) -extender;
- c. $f[\kappa] \subseteq H_\kappa$;
- d. $f \restriction \lambda = j_E(f \restriction \kappa) \restriction \lambda (= j_E(f) \restriction \lambda)$.

II. If δ is Woodin, then

$$(\forall f : \delta \rightarrow H_\delta)(\exists \kappa < \delta)(\forall \lambda \in (\kappa, \delta) \cap \text{Lim})(\exists E) \text{ strong}(E, \kappa, \lambda, f).$$

III. **Definition.** Suppose that:

- a. δ is inaccessible;
- b. $\mathbf{P} \subseteq H_\delta$ a set of propositional letters;
- c. $f : \delta \rightarrow \mathcal{L}_\delta(\mathbf{P})$;
- d. $\kappa < \lambda < \delta$.

Then the formula $\Psi_{f,\kappa,\lambda}$ of $\mathcal{L}_\delta(\mathbf{P})$ is defined to be

$$\bigwedge_{\xi < \kappa} f(\xi) \implies \bigwedge_{\xi < \lambda} f(\xi).$$

IV. **Definition.** Suppose that:

- a. δ is inaccessible;
- b. $\mathbf{P} \subseteq H_\delta$ is a set of propositional letters;

c. $\mathcal{E} \subseteq V_\delta$ is a set of extenders.

Then

- i. $\text{ext_thr}(\delta, \mathbf{P}, \mathcal{E}) := \{\Psi_{f, \kappa, \lambda} : (\exists E \in \mathcal{E}) \text{ strong}(E, \kappa, \lambda, f)\};$
- ii. $\mathbb{W}(\delta, \mathbf{P}, \mathcal{E}) := \mathbb{B}_{\delta, \mathbf{P}}(\text{ext_thr}(\delta, \mathbf{P}, \mathcal{E}))$, the so-called extender algebra.

V. $\text{ext_thr}(\delta, \mathbf{P}, \mathcal{E})$ is consistent.

- a. We show that $\mathbf{0}_{\mathbf{P}} \models \text{ext_thr}(\delta, \mathbf{P}, \mathcal{E})$.
- b. Let $E \in \mathcal{E}$ be arbitrary satisfying $\text{strong}(E, \kappa, \lambda, f)$. We want to show that

$$\mathbf{0}_{\mathbf{P}} \models \bigwedge_{\xi < \kappa} f(\xi) \implies \bigwedge_{\xi < \lambda} f(\xi).$$

- c. Assume that $\mathbf{0}_{\mathbf{P}} \models \bigwedge_{\xi < \kappa} f(\xi)$.
- d. This implies $\mathbf{0}_{j_E(\mathbf{P})} \models \bigwedge_{\xi < j_E(\kappa)} j_E(f)(\xi)$ (within $\mathcal{L}_\delta(j_E(\mathbf{P}))$).
- e. Thus, $\mathbf{0}_{j_E(\mathbf{P})} \models \bigwedge_{\xi < \lambda} j_E(f)(\xi)$.
- f. Since $j_E(f) \upharpoonright \lambda = f \upharpoonright \lambda$, we have that $\mathbf{0}_{j_E(\mathbf{P})} \models \bigwedge_{\xi < \lambda} f(\xi)$, as well as that the relevant portion of $\mathbf{P}$ is fixed under j_E .
- g. We conclude $\mathbf{0}_{\mathbf{P}} \models \bigwedge_{\xi < \lambda} f(\xi)$.

VI. Suppose that $|\mathbf{P}| < \min\{\text{crit}(E) : E \in \mathcal{E}\}$. Then for every $x : \mathbf{P} \rightarrow 2$, we have $x \models \text{ext_thr}(\delta, \mathbf{P}, \mathcal{E})$.

Proof.

- 1° We may assume that $\gamma := \min\{\text{crit}(E) : E \in \mathcal{E}\}$ is defined.
- 2° We may assume w.l.o.g. that $\mathbf{P} \in V_\gamma$.
- 3° Let $E \in \mathcal{E}$ be arbitrary satisfying $\text{strong}(E, \kappa, \lambda, f)$. We want to show that

$$x \models \bigwedge_{\xi < \kappa} f(\xi) \implies \bigwedge_{\xi < \lambda} f(\xi).$$

4° Assume that $x \models \bigwedge_{\xi < \kappa} f(\xi)$.

5° This implies

$$\text{Ult}(V, E) \models \left(j_E(x) \models \bigwedge_{\xi < j_E(\kappa)} j_E(f)(\xi) \right).$$

6° Since $j_E(x) = x$ & $\lambda \leq j_E(\kappa)$ & $j_E(f) \upharpoonright \lambda = f \upharpoonright \lambda$, we conclude

$$x \models \bigwedge_{\xi < \lambda} f(\xi).$$

□

VII. Suppose that:

- a. $\mathbb{P}$ is a poset;
- b. $|\mathbb{P}| < \min\{\text{crit}(E) : E \in \mathcal{E}\}$;
- c. $|\mathbb{P}| < \delta$;
- d. $|\mathbb{P}| \leq |\mathbf{P}|$.

Then there is $[\phi] > 0$ and a complete embedding $i : \mathbb{P} \rightarrow \mathbb{W}(\delta, \mathbf{P}, \mathcal{E}) \restriction [\phi]$.

Proof.

- 1° We may assume w.l.o.g. that $\mathbb{P} = \gamma \leq |\mathbf{P}|$.
- 2° Let $(p_\xi : \xi < \gamma)$ be distinct elements of $\mathbf{P}$. We may assume w.l.o.g. that they are all below $\min\{\text{crit}(E) : E \in \mathcal{E}\}$
- 3° Let ϕ be the conjunction of the following formulas:
 - a. $p_\xi \implies p_\eta$, for every $\xi \leq_{\mathbb{P}} \eta$;
 - b. $\neg(p_\xi \wedge p_\eta)$, for every $\xi \perp_{\mathbb{P}} \eta$;
 - c. $\bigvee_{\xi \in D} p_\xi$, for every predense $D \subseteq \mathbb{P}$.
- 4° ϕ is consistent with $\text{ext_thr}(\delta, \mathbf{P}, \mathcal{E})$.

Proof.

- 1' We may assume w.l.o.g. that $\mathbb{P}$ is below $\min\{\text{crit}(E) : E \in \mathcal{E}\}$.
- 2' Let $G \rightsquigarrow V^{\mathbb{P}}$ be arbitrary.
- 3' Let $x : \mathbf{P} \rightarrow 2$ be defined by

$$x(p) := \begin{cases} 1, & (\exists \xi < \gamma)(p = p_\xi \wedge \xi \in G) \\ 0, & \text{otherwise.} \end{cases}$$

- 4' It is clear that $x \models \phi$.
- 5' Let us show that $x \models \text{ext_thr}(\delta, \mathbf{P}, \mathcal{E})$.
- 6' Let $E \in \mathcal{E}$ be arbitrary satisfying $\text{strong}(E, \kappa, \lambda, f)$. We want to show that

$$x \models \bigwedge_{\xi < \kappa} f(\xi) \implies \bigwedge_{\xi < \lambda} f(\xi).$$

- 7' Assume that $x \models \bigwedge_{\xi < \kappa} f(\xi)$.
- 8' Since $\mathbb{P} \in V \restriction \text{crit}(j_E)$, there $j : V[G] \prec \text{Ult}(V, E)[G]$ such that $j \restriction V = j_E^V$.
- 9' Since $j(f) \restriction \lambda = j_E(f) \restriction \lambda = f \restriction \lambda$, we conclude that the part of x relevant for evaluating $\bigwedge_{\xi < \lambda} f(\xi)$ is fixed under j .

10' By the elementarity, we have

$$\text{Ult}(V, E)[G] \models \left(j(x) \models \bigwedge_{\xi < j(\kappa)} j(f)(\xi) \right).$$

11' Since $\lambda \leq j(\kappa)$ & $j(f) \restriction \lambda = f \restriction \lambda$ & the relevant part of x is fixed under j , we conclude

$$x \models \bigwedge_{\xi < \lambda} f(\xi).$$

□

5° Let $i : \mathbb{P} \rightarrow \mathbb{W}(\delta, \mathbf{P}, \mathcal{E}) \restriction [\phi]$ be defined by $i(\xi) := [p_\xi \wedge \phi]$.

6° i is a complete embedding.

Proof.

1' For $\xi \leq_{\mathbb{P}} \eta$, we have $\phi \vdash p_\xi \implies p_\eta$, which implies $\vdash p_\xi \wedge \phi \implies p_\eta \wedge \phi$ and $i(\xi) \leq i(\eta)$.

2' For $\xi \perp_{\mathbb{P}} \eta$, we have $\phi \vdash \neg(p_\xi \wedge p_\eta)$, which implies $\vdash \neg((p_\xi \wedge \phi) \wedge (p_\eta \wedge \phi))$ and $i(\xi) \cdot i(\eta) = 0$.

3' For $D \subseteq \gamma$ predense, we have $\phi \vdash \bigwedge_{\xi \in D} p_\xi$, so $[\phi] = \sum_{\xi \in D} i(\xi)$.

□

□

VIII. Suppose that $E \in \mathcal{E}$ satisfies **strong**(E, κ, λ, f). Then

$$\text{ext_thr}(\delta, \mathbf{P}, \mathcal{E}) \vdash \bigwedge_{\xi < \kappa} f(\xi) \implies f(\kappa).$$

Proof.

1° $\text{ext_thr}(\delta, \mathbf{P}, \mathcal{E}) \vdash \bigwedge_{\xi < \kappa} f(\xi) \implies \bigwedge_{\xi < \lambda} f(\xi)$

2° $\vdash \bigwedge_{\xi < \lambda} f(\xi) \implies f(\kappa)$

3° $\text{ext_thr}(\delta, \mathbf{P}, \mathcal{E}) \vdash \bigwedge_{\xi < \kappa} f(\xi) \implies f(\kappa)$

□

IX. Suppose that:

- a. δ is Woodin;
- b. $\mathcal{E}$ is the set of all E satisfying:
 - i. E is a strong (κ, λ) -extender;
 - ii. λ is inaccessible.

Then $\mathbb{W}(\delta, \mathbf{P}, \mathcal{E})$ has δ -c.c. In particular, $\mathbb{W}(\delta, \mathbf{P}, \mathcal{E})$ is a complete boolean algebra

Proof.

- 1° Let $([\phi_\xi] : \xi < \delta)$ be an arbitrary family of non-zero elements. We want to show that it is not an anti-chain.
- 2° Let $f := (\neg\phi_\xi : \xi < \delta)$.
- 3° Since δ is Woodin, there is E satisfying:
 - a. E is a strong (κ, λ) -extender;
 - b. λ is inaccessible;
 - c. $f[\kappa] \subseteq H_\kappa$;
 - d. $f \restriction \lambda = j_E(f) \restriction \lambda$.
- 4° In other words, $E \in \mathcal{E}$ and $\text{strong}(E, \kappa, \lambda, f)$.
- 5° This means that

$$\text{ext_thr}(\delta, \mathbf{P}, \mathcal{E}) \vdash \bigwedge_{\xi < \kappa} \neg\phi_\xi \implies \neg\phi_\kappa.$$

- 6° Equivalently,

$$\text{ext_thr}(\delta, \mathbf{P}, \mathcal{E}) \vdash \phi_\kappa \implies \bigvee_{\xi < \kappa} \phi_\xi.$$

- 7° Equivalently,

$$[\phi_\kappa] \leq \sum_{\xi < \kappa} [\phi_\xi].$$

- 8° Since $[\phi_\kappa] > 0$, this implies that for some $\xi < \kappa$, we have $[\phi_\kappa] \cdot [\phi_\xi] > 0$.

□

4 Genericity Iterations

I. Suppose that:

- a. M is a countable transitive model of set theory;
- b. $[\text{in } M] \delta$ is Woodin;
- c. $[\text{in } M] \mathcal{E}$ is the set of all E satisfying:
 - i. E is a strong (κ, λ) -extender;
 - ii. λ is inaccessible;
- d. M is $(\omega_1 + 1)$ - $\mathcal{E}$ -iterable;
- e. $x \subseteq \omega$.

Then there is an $\mathcal{E}$ -iteration tree $\mathcal{T}$ of length $\alpha + 1 < \omega_1$ on M such that x corresponds to an $M_\alpha^\mathcal{T}$ -generic filter on $j_{0,\alpha}^\mathcal{T}(\mathbb{W}(\delta, \omega, \mathcal{E})^M)$.

Proof.

- 1° Let Σ be a winning strategy for II in the $\mathcal{E}$ -iteration game on M of length $(\omega_1 + 1)$.
- 2° Let us define an iteration tree $\mathcal{T}$ on M of length $\leq \omega_1$, by recursion.
- 3° Suppose that $\mathcal{T} \upharpoonright (\alpha + 1)$ has been defined.
- 4° If $x \models \mathbb{W}(j_{0,\alpha}(\delta), \omega, j_{0,\alpha}(\mathcal{E}))^{M_\alpha}$, we halt the construction. (Recall that this means that x is generic for the extender algebra over M_α .)
- 5° Otherwise, there are $E \in \mathcal{E}, \kappa, \lambda, f$ such that $\text{strong}(E, \kappa, \lambda, f)$ and $x \not\models \Psi_{f, \kappa, \lambda}$.
- 6° We set E_α to be one such E .
- 7° At the limit stages of the construction, we choose branches according to Σ .
- 8° This concludes the construction.
- 9° If the length of $\mathcal{T}$ is $< \omega_1$, then $\mathcal{T}$ is as required.
- 10° In what follows, we show that the case where the length of $\mathcal{T}$ is ω_1 , is not possible.
- 11° Assume otherwise.
- 12° Let $b := \Sigma(\mathcal{T})$, a well-founded brunch, and let $M_{\omega_1} := M_b^\mathcal{T}$. We denote this tree of length $\omega_1 + 1$ by $\mathcal{T}'$.
- 13° Let $k : (H, \in, \mathcal{U}, \alpha, c, x) \prec (H_{\omega_2}, \in, \mathcal{T}', \omega_1, b, x)$ be a countable hull.
- 14° By lemma 2-I, we have that
 - a. $\mathcal{U} = \mathcal{T} \upharpoonright (\alpha + 1)$,
 - b. $c = b \cap \alpha$,
 - c. $\alpha \in b$,
 - d. $k \upharpoonright M_\alpha = j_{\alpha, \omega_1}$.
- 15° The extender applied to M_α along b is equal to E_β for some $\beta \geq \alpha$.
- 16° We have $\text{crit}(E_\beta) = \text{crit}(j_{\alpha, \beta}) = \text{crit}(j_{\alpha, \omega_1}) = \text{crit}(k) = \alpha$. Let us denote the length of E_β by λ .
- 17° Let $f : j_{0, \beta}(\delta) \rightarrow \mathcal{L}_{j_{0, \beta}(\delta)}(\omega)^{M_\beta}$ be a function in M_β such that:
 - a. $f[\alpha] \subseteq \mathcal{L}_\alpha(\omega)^{M_\beta}$;
 - b. $f \upharpoonright \lambda = j_{E_\beta}^{M_\beta}(f) \upharpoonright \lambda$;
 - c. $x \models \bigwedge_{\xi < \alpha} f(\xi)$;
 - d. $x \not\models \bigwedge_{\xi < \lambda} f(\xi)$.
- 18° $M_\alpha \upharpoonright (\alpha + 1) = M_\beta \upharpoonright (\alpha + 1)$

- 19° $f \restriction \alpha$ can be coded as an element of $M_\beta \restriction (\alpha + 1)$.
 20° In particular, $f \restriction \alpha \in M_\alpha \restriction (\alpha + 1)$.
 21° $[\text{In } k(M_\alpha)]$ $k(f \restriction \alpha)$ is an ω_1^V -sequence of formulas such that

$$x \models \bigwedge_{\xi < \omega_1} k(f \restriction \alpha)(\xi).$$

- 22° $[\text{In } k(M_\alpha)]$ $x \models \bigwedge_{\xi < \lambda} k(f \restriction \alpha)(\xi)$.
 23° Note now that $k(f \restriction \alpha) \restriction \lambda = j_{\alpha, \omega_1}(f \restriction \alpha) \restriction \lambda = j_{\alpha, \beta+1}(f \restriction \alpha) \restriction \lambda = j_{E_\beta}^{M_\alpha}(f \restriction \alpha) \restriction \lambda = j_{E_\beta}^{M_\beta}(f \restriction \alpha) \restriction \lambda = f \restriction \lambda$. (The penultimate equality follows from 19°.)
 24° The previous two points now yield $x \models \bigwedge_{\xi < \lambda} f(\xi)$. This contradicts 17°d.

□